

Coronal Heating as Bridge Dissipation on $X(2)$: Why the Solar Corona Is Hotter Than the Surface

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Abstract

The solar coronal heating problem—why the corona at $T \sim 10^6$ K is two orders of magnitude hotter than the photosphere at $T \sim 6 \times 10^3$ K—has resisted a definitive resolution for over 70 years. We show that within the LoNalogy framework, the temperature inversion is a one-line theorem: the per-particle heating $q_{\text{br}}(s) = Q_{\text{br}}(s)/n(s)$ is monotonically increasing toward the bridge peak because the sech^2 bridge dissipation profile is divided by an exponentially decreasing density. Concretely, the bridge heating density $Q_{\text{br}}(s) = Q_0 \text{sech}^2((s - s_B)/\ell_B)$ and the hydrostatic density $n(s) = n_0 e^{-s/H}$ yield $d(\log q_{\text{br}})/ds > 0$ for $s < s_B$. A steady-state energy balance then forces a coronal temperature maximum above the photosphere. The 70-year debate between wave heating and nanoflare/reconnection heating is resolved by showing that both are basis decompositions of the same positive bridge quadratic form $Q_0 = \eta \langle J, \Delta_{\text{br}} J \rangle$.

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1 Introduction

The solar corona presents a fundamental paradox: it is two orders of magnitude hotter than the underlying photosphere, despite being further from the energy source. Since Grotrian (1939) and Edlén (1943) identified coronal emission lines as highly ionised iron, the question “why is the corona hot?” has been a central problem in solar physics.

The two main candidate mechanisms are:

- (i) **Wave heating:** Alfvén waves generated at the photosphere propagate upward and dissipate in the corona via turbulent cascade or phase mixing.
- (ii) **Nanoflare/reconnection heating:** magnetic field lines braided by photospheric convection develop current sheets that dissipate via magnetic reconnection.

Recent observations have converged on a mixed picture: closed-field active regions are predominantly heated by braiding/reconnection, while open-field coronal holes are heated by Alfvénic turbulence and switchbacks.

In this paper, we show that the LoNalogy framework provides a unified resolution. The three-sector decomposition $\mathfrak{U} = \mathfrak{I} \oplus \mathfrak{V} \oplus \mathfrak{B}$ maps onto the solar atmosphere as visible plasma $\oplus$ magnetic bridge $\oplus$ convective hidden sector. The sech^2 bridge weight, already derived in the core framework from $X(2)$ modular geometry, provides the spatial profile of magnetic dissipation. Combined with the exponential density stratification, this yields the temperature inversion as a one-line inequality.

2 Solar Dictionary

Definition 2.1 (Three-sector solar decomposition). The solar atmosphere is mapped to the LoNalogy three-sector structure as:

$$\mathfrak{U}_{\odot} = \mathfrak{V}_{\text{ph}} \oplus \mathfrak{B}_{\text{mag}} \oplus \mathfrak{I}_{\text{conv}}, \quad (2.1)$$

where:

- $\mathfrak{V}_{\text{ph}}$: the optically visible photosphere/chromosphere plasma (observable sector),
- $\mathfrak{B}_{\text{mag}}$: coronal loops, current sheets, and Alfvén channels (magnetic bridge),
- $\mathfrak{I}_{\text{conv}}$: convective zone footpoint motions and buried topology stress (hidden sector).

Definition 2.2 (Bridge frame operator). The magnetic bridge carries a positive frame operator

$$\Delta_{\text{br}} = \Sigma^\dagger \Sigma \geq A_{\text{br}} I > 0, \quad (2.2)$$

where Σ is the bridge embedding and $A_{\text{br}} > 0$ is the bridge floor constant. For footpoint driving current $J \neq 0$, the bridge power amplitude is

$$Q_0 := \eta \langle J, \Delta_{\text{br}} J \rangle > 0, \quad (2.3)$$

where $\eta > 0$ is a dissipation efficiency. The positivity $Q_0 > 0$ is automatic from the positivity of Δ_{br} .

Remark 2.3. The statement “magnetic fields are the energy source of coronal heating” translates in LoNalogy to “the bridge quadratic form is positive definite.” This is not an assumption but a consequence of the frame operator structure $\Delta_{\text{br}} = \Sigma^\dagger \Sigma$.

3 Bridge Heating Profile

Definition 3.1 (Bridge heating density). Let s be the coordinate along a magnetic field line, with $s = 0$ at the photosphere and s increasing into the corona. The bridge peak is located at $s = s_B > 0$. The bridge deviation variable is

$$\Theta(s) := \frac{s - s_B}{\ell_B}, \quad (3.1)$$

where $\ell_B > 0$ is the bridge scale length. The bridge heating density (energy per unit volume per unit time) is

$$Q_{\text{br}}(s) = Q_0 \operatorname{sech}^2\left(\frac{s - s_B}{\ell_B}\right). \quad (3.2)$$

This follows from the LoNalogy Fisher weight $\mathcal{F}(s) = \operatorname{sech}^2 \Theta(s)$ applied to the solar bridge.

Definition 3.2 (Hydrostatic density stratification). The plasma number density along the field line is approximated by hydrostatic equilibrium:

$$n(s) = n_0 e^{-s/H}, \quad (3.3)$$

where n_0 is the photospheric density and H is the pressure scale height ($H \approx 50\text{--}100$ Mm in the corona).

4 Coronal Inversion Theorem

Theorem 4.1 (Coronal inversion). *The per-particle bridge heating is*

$$q_{\text{br}}(s) := \frac{Q_{\text{br}}(s)}{n(s)} = \frac{Q_0}{n_0} e^{s/H} \operatorname{sech}^2\left(\frac{s - s_B}{\ell_B}\right), \quad (4.1)$$

and its logarithmic derivative is

$$\frac{d}{ds} \log q_{\text{br}}(s) = \frac{1}{H} - \frac{2}{\ell_B} \tanh\left(\frac{s - s_B}{\ell_B}\right). \quad (4.2)$$

For $0 \leq s < s_B$, the tanh term is negative, so

$$\frac{d}{ds} \log q_{\text{br}}(s) > 0 \quad \text{for } 0 \leq s < s_B. \quad (4.3)$$

That is, the per-particle heating is monotonically increasing from the photosphere toward the bridge peak.

Proof. Step 1: Compute q_{br} . By (3.2) and (3.3):

$$q_{\text{br}}(s) = \frac{Q_0 \operatorname{sech}^2((s - s_B)/\ell_B)}{n_0 e^{-s/H}} = \frac{Q_0}{n_0} e^{s/H} \operatorname{sech}^2\left(\frac{s - s_B}{\ell_B}\right). \quad (4.4)$$

Step 2: Take the logarithmic derivative.

$$\log q_{\text{br}}(s) = \log \frac{Q_0}{n_0} + \frac{s}{H} + \log \operatorname{sech}^2\left(\frac{s - s_B}{\ell_B}\right). \quad (4.5)$$

Differentiating term by term:

$$\frac{d}{ds} \log q_{\text{br}} = \frac{1}{H} + \frac{d}{ds} \left[-2 \log \cosh\left(\frac{s - s_B}{\ell_B}\right) \right] \quad (4.6)$$

$$= \frac{1}{H} - \frac{2}{\ell_B} \tanh\left(\frac{s - s_B}{\ell_B}\right). \quad (4.7)$$

Step 3: Sign analysis. For $s < s_B$: $(s - s_B)/\ell_B < 0$, so $\tanh((s - s_B)/\ell_B) < 0$. Therefore the second term is positive, and both terms contribute positively:

$$\frac{d}{ds} \log q_{\text{br}}(s) = \underbrace{\frac{1}{H}}_{>0} - \underbrace{\frac{2}{\ell_B} \tanh\left(\frac{s - s_B}{\ell_B}\right)}_{>0 \text{ for } s < s_B} > 0. \quad (4.8) \quad \square$$

Remark 4.2 (Physical interpretation). The two terms have distinct physical origins:

- $1/H > 0$: the exponential density decrease means fewer particles share the available energy at higher altitudes.
- $-(2/\ell_B) \tanh \Theta > 0$ (for $s < s_B$): the sech^2 profile concentrates heating toward the bridge peak.

Both effects cooperate below the bridge peak: the heating increases while the density decreases, driving the per-particle heating monotonically upward.

5 Steady-State Temperature Maximum

Theorem 5.1 (Coronal temperature maximum). *Consider a coronal loop of half-length L with the steady-state energy balance equation along the field line:*

$$\frac{dF_c}{ds} = Q_{\text{br}}(s) - n(s)^2 \Lambda(T(s)), \quad (5.1)$$

where $F_c := -\kappa_{\parallel} T^{5/2} dT/ds$ is the Spitzer conductive heat flux and $\Lambda(T)$ is the optically thin radiative loss function. Impose the symmetry (apex) condition $F_c(L) = 0$.

If the net bridge heating exceeds the radiative losses integrated over the loop:

$$\int_0^L (Q_{\text{br}}(s) - n(s)^2 \Lambda(T(s))) ds > 0, \quad (5.2)$$

then there exists $s_* \in (0, L)$ such that

$$\boxed{T(s_*) > T(0) = T_{\text{ph}}.} \quad (5.3)$$

That is, the corona is necessarily hotter than the photosphere.

Proof. Step 1: Integrate the energy equation. Integrating (5.1) from 0 to L :

$$F_c(L) - F_c(0) = \int_0^L (Q_{\text{br}}(s) - n(s)^2 \Lambda(T(s))) ds. \quad (5.4)$$

By the apex condition $F_c(L) = 0$:

$$F_c(0) = - \int_0^L (Q_{\text{br}}(s) - n(s)^2 \Lambda(T(s))) ds. \quad (5.5)$$

Step 2: Sign of the base heat flux. By hypothesis (5.2), the integral is positive, so $F_c(0) < 0$.

Step 3: Direction of heat flow. $F_c(0) < 0$ means $-\kappa_{\parallel} T^{5/2} (dT/ds)|_{s=0} < 0$, i.e., $dT/ds|_{s=0} > 0$. Furthermore, $F_c(0) < 0$ means that heat flows *downward* (from corona to photosphere) at the base.

Step 4: Existence of temperature maximum. For heat to flow downward at the base, there must exist a region above the base where the temperature is higher than T_{ph} . Since $F_c(L) = 0$ (no heat flux at the apex) and $F_c(0) < 0$ (downward flux at the base), the temperature must have a maximum somewhere in $(0, L)$. Formally, if $T(s) \leq T_{\text{ph}}$ for all $s \in [0, L]$, then $dT/ds \leq 0$ everywhere, giving $F_c \geq 0$ everywhere (heat would flow upward or not at all), contradicting $F_c(0) < 0$. $\square$

Corollary 5.2 (Why hot?). *The corona is hot because:*

- (i) *Bridge dissipation $Q_{\text{br}}(s) = Q_0 \text{sech}^2 \Theta(s)$ deposits energy along coronal magnetic structures.*
- (ii) *The density $n(s) = n_0 e^{-s/H}$ drops exponentially, so per-particle heating increases with altitude.*
- (iii) *Net bridge heating exceeds radiative losses, forcing the conductive heat flux to point downward from corona to surface.*
- (iv) *This requires a temperature maximum above the surface.*

$\boxed{\text{Corona is hot because low-density magnetic bridge dissipation exceeds radiative losses.}}$

(5.6)

6 Unification of Waves and Nanoflares

Theorem 6.1 (Basis independence of bridge heating). *The bridge power amplitude*

$$Q_0 = \eta \langle J, \Delta_{\text{br}} J \rangle \quad (6.1)$$

is independent of the choice of basis for the bridge space $\mathfrak{B}_{\text{mag}}$. In particular:

- (i) *In the **current-sheet basis** (eigenbasis of $\nabla \times \mathbf{B}$), Q_0 decomposes into reconnection events (nanoflares).*
- (ii) *In the **Alfvénic basis** (eigenbasis of the Alfvén wave operator), Q_0 decomposes into wave damping amplitudes.*
- (iii) *Both decompositions yield the same total heating Q_0 .*

Proof. $\Delta_{\text{br}} = \Sigma^\dagger \Sigma$ is a positive self-adjoint operator. The quadratic form $\langle J, \Delta_{\text{br}} J \rangle$ is basis-independent by the spectral theorem: for any unitary U ,

$$\langle UJ, U\Delta_{\text{br}}U^\dagger UJ \rangle = \langle J, \Delta_{\text{br}} J \rangle. \quad (6.2)$$

Therefore, the total heating amplitude is the same whether computed in the current-sheet basis or the Alfvénic basis. The “wave vs. nanoflare” distinction is a choice of orthonormal basis for $\mathfrak{B}_{\text{mag}}$, not a physical difference. $\square$

Remark 6.2 (Resolution of the 70-year debate). The LoNalogy answer to “waves or nanoflares?” is: *both are basis decompositions of the same bridge quadratic form.* Observationally:

- Closed-field active regions: the reconnection/braiding basis is observationally natural (current sheets are resolved).
- Open-field coronal holes: the Alfvénic basis is observationally natural (switchbacks and wave amplitudes are measured).

The total heating Q_0 is the same in both bases. The apparent competition between the two mechanisms is an artefact of basis choice.

7 Why Hot but Dim?

Proposition 7.1 (Hot but dim). *The corona is simultaneously hotter and dimmer than the photosphere. In LoNalogy, this is not a paradox but a direct consequence of the density dependence.*

Temperature is controlled by per-particle heating:

$$T \sim q_{\text{br}} \sim \frac{Q_{\text{br}}}{n} \sim \frac{Q_0 \text{sech}^2 \Theta}{n_0 e^{-s/H}}. \quad (7.1)$$

As n decreases (upward), T increases.

Brightness (*optically thin EUV/X-ray emissivity*) is controlled by:

$$\epsilon \sim n^2 G(T), \quad (7.2)$$

where $G(T)$ is the contribution function. As n decreases, $\epsilon \propto n^2$ drops quadratically.

Therefore:

$$\boxed{n \downarrow \implies T \uparrow, \quad \epsilon \propto n^2 \downarrow. \quad \text{Hot but dim.}} \quad (7.3)$$

Proof. The temperature scaling follows from Theorem 4.1: q_{br} increases with altitude below the bridge peak. The emissivity scaling $\epsilon \propto n^2$ is standard for optically thin thermal bremsstrahlung and line emission. Since $n \propto e^{-s/H}$, $\epsilon \propto e^{-2s/H}$ decreases faster than $q_{\text{br}} \propto e^{s/H} \text{sech}^2 \Theta$ increases (for $s < s_B$). $\square$

8 Compact Summary

The complete LoNalogy coronal heating closure is:

Three-sector decomposition:

$$\mathfrak{U}_{\odot} = \mathfrak{V}_{\text{ph}} \oplus \mathfrak{B}_{\text{mag}} \oplus \mathfrak{J}_{\text{conv}}, \quad \Delta_{\text{br}} = \Sigma^{\dagger} \Sigma \geq A_{\text{br}} I. \quad (8.1)$$

Bridge heating profile:

$$Q_{\text{br}}(s) = Q_0 \text{sech}^2\left(\frac{s - s_B}{\ell_B}\right), \quad n(s) = n_0 e^{-s/H}. \quad (8.2)$$

Per-particle heating monotonicity:

$$\frac{d}{ds} \log \frac{Q_{\text{br}}(s)}{n(s)} = \frac{1}{H} - \frac{2}{\ell_B} \tanh\left(\frac{s - s_B}{\ell_B}\right) > 0 \quad (s < s_B). \quad (8.3)$$

Temperature maximum:

$$\int_0^L (Q_{\text{br}} - n^2 \Lambda) ds > 0 \implies T_{\text{cor}} > T_{\text{ph}}. \quad (8.4)$$

Basis unification:

$$Q_0 = \eta \langle J, \Delta_{\text{br}} J \rangle \quad (\text{basis-independent}). \quad (8.5)$$

In one sentence:

The solar corona is hot because sech^2 bridge dissipation is divided by exponentially decreasing density, and the corona is dim because emissivity scales as n^2 .

9 What Is Closed and What Remains

Result	Reference	Status
Per-particle heating monotonicity	Theorem 4.1	Closed
Coronal temperature maximum	Theorem 5.1	Closed
Wave/nanoflare unification	Theorem 6.1	Closed
Hot but dim	Proposition 7.1	Closed
Absolute heating amplitude Q_0	Eq. (2.3)	Open (calibration)
Loop geometry (s_B, ℓ_B, L)	—	Open (observation)
Footpoint current J reconstruction	—	Open (magnetogram)

The conceptual core—*why* the temperature inversion occurs—is fully closed. What remains is the calibration of specific active regions and coronal holes against observational data.

Remark 9.1 (Extension layer). This paper is an *extension layer* of the LoNalogy framework. The core theorems (Schur reduction, bridge weight sech^2 , positive frame operator) are rigorous. The solar dictionary (Definition 2.1) is an interpretive mapping. The strength of the result is that the *qualitative* conclusion (temperature inversion) follows from the *sign* of the inequality (4.3), which is robust to the specific form of the dictionary.

Conclusion

The solar coronal heating problem reduces in LoNalogy to a one-line inequality:

$$\frac{d}{ds} \log \frac{Q_{\text{br}}(s)}{n(s)} > 0 \quad (s < s_B). \quad (9.1)$$

The sech^2 bridge dissipation profile, divided by the exponential density stratification, yields monotonically increasing per-particle heating toward the bridge peak. A steady-state energy balance then forces a coronal temperature maximum. The wave-versus-nanoflare debate is dissolved: both are basis decompositions of the same positive bridge quadratic form.

The corona is not hot despite being far from the energy source. It is hot *because* it is on the magnetic bridge, where hidden convective energy dissipates into a low-density plasma.

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